$$f(x) = \log_{\frac{1}{2}} \left(\frac{x^2 - x - 2}{x^2 - 3} \right)$$

STUDIARE DOMINIO E POSITIVITÀ

$$\frac{x^2 - x - 2}{x^2 - 3} > 0 \qquad \left(\begin{array}{l} x^2 - 3 \neq 0 \\ \hline \left[\begin{array}{l} x^2 - 3 > 0 \end{array} \right. \end{array} \right.$$

$$x^2 - x - 2 > 0$$

$$x^2 - x - 2 = 0$$

$$x_{1,2} = \frac{1 \pm \sqrt{1 + 8}}{2} = \frac{1 \pm 3}{2} \begin{cases} -1 \\ 2 \end{cases}$$

$$x = -1 \quad , \quad x = 2$$

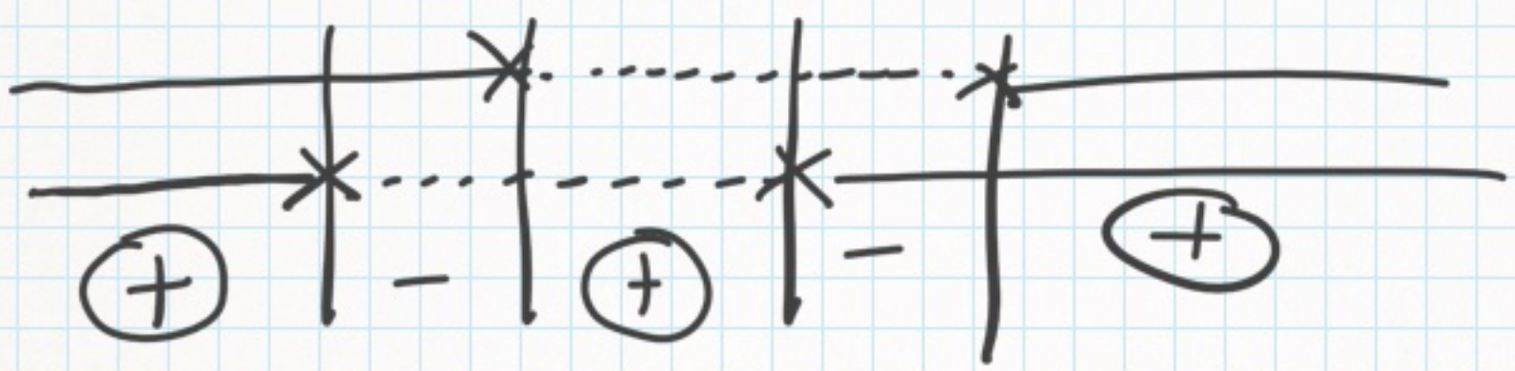
$$x^2 - x - 2 > 0 \iff x < -1 \vee x > 2$$

$x^2 - 3 > 0 \iff x < -\sqrt{3} \vee x > \sqrt{3}$

$x^2 - 3 = 0 \iff x = \pm\sqrt{3}$

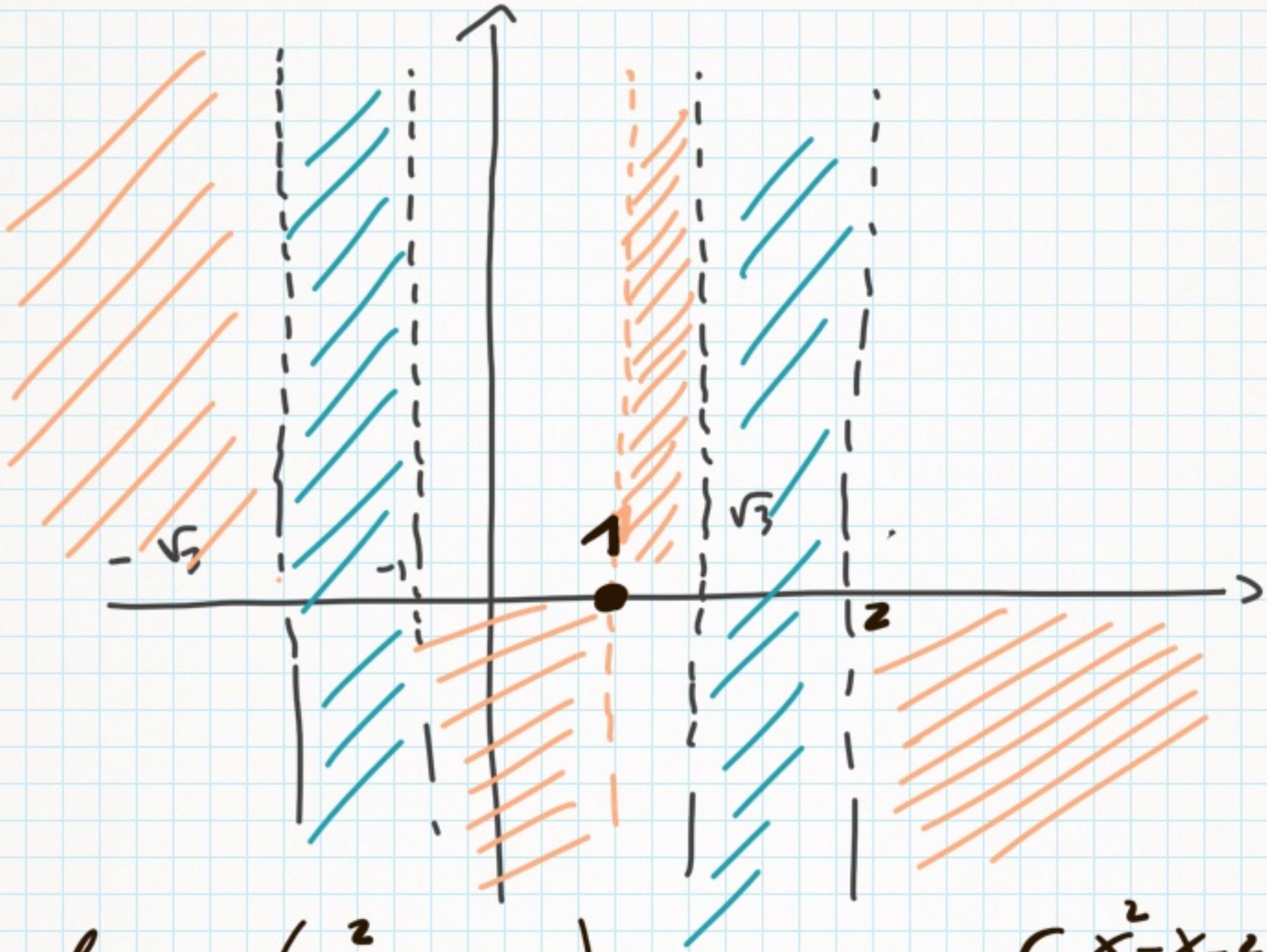
$$\frac{x^2 - x - 2}{x^2 - 3} > 0$$

$-\sqrt{3} \quad -1 \quad \sqrt{3} \quad 2$



Dominio:

$x < -\sqrt{3} \vee -1 < x < \sqrt{3} \vee x > 2$



$$\log_{\frac{1}{2}}\left(\frac{x^2-x-2}{x^2-3}\right)\geq 0$$

$$0<\frac{x^2-x-2}{x^2-3}\leq 1$$

$$\begin{cases}\dfrac{x^2-x-2}{x^2-3}>0\\[2ex]\dfrac{x^2-x-2}{x^2-3}\leq 1\end{cases}$$

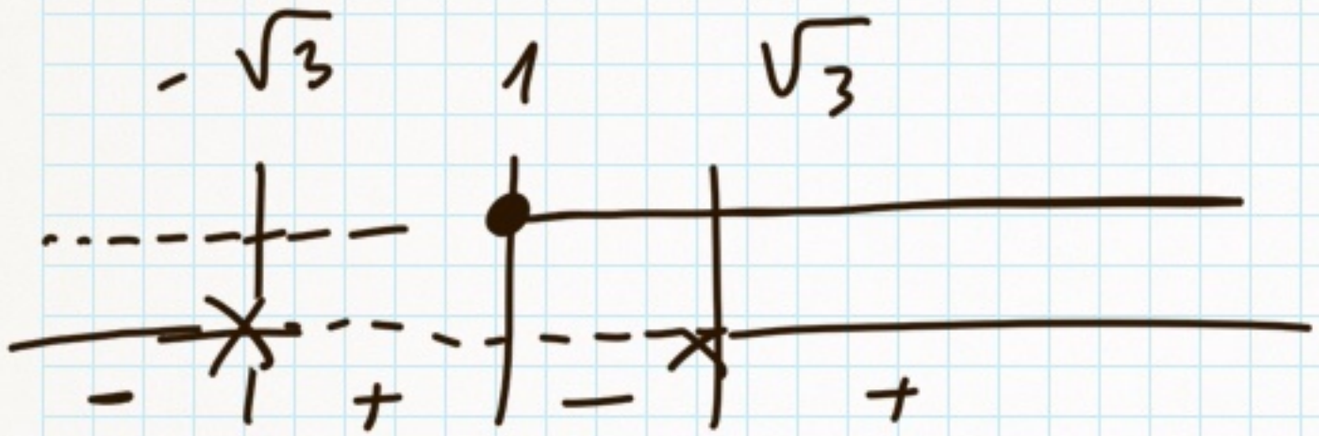
$$\begin{cases} x < -\sqrt{3} \vee -1 < x < \sqrt{3} \vee x > 2 \\ \dfrac{x^2 - x - 2}{x^2 - 3} - 1 \leq 0 \end{cases}$$

$$\frac{\cancel{x^2} - x - 2 - \cancel{x^2} + 3}{x^2 - 3} \leq 0$$

$$\frac{-x + 1}{x^2 - 3} \leq 0 \qquad \text{(moltiplico per } -1\text{)}$$

$$\frac{x - 1}{x^2 - 3} \geq 0$$

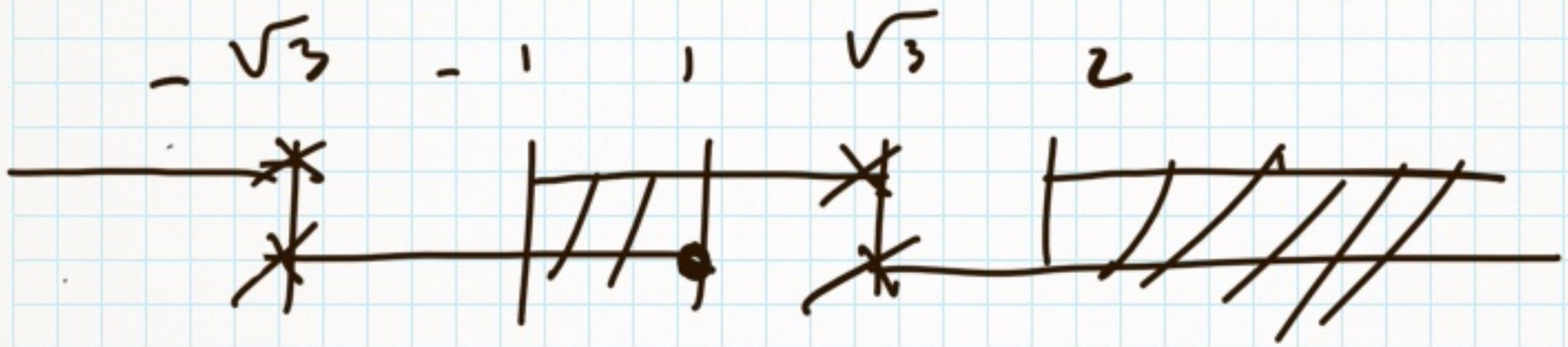
$$\frac{x - 1 \geq 0}{x^2 - 3 > 0} \qquad \begin{matrix} x \geq 1 \\ x < -\sqrt{3} \vee x > \sqrt{3} \end{matrix}$$



$-\sqrt{3} < x \leq 1 \quad \vee \quad x > \sqrt{3}$

Adesso risolviamo il sistema

$$\begin{cases} x < -\sqrt{3} \vee -1 < x < \sqrt{3} \vee x > 2 \\ -\sqrt{3} < x \leq 1 \vee x > \sqrt{3} \end{cases}$$



$f(x) \geq 0 \iff -1 < x \leq 1 \quad \vee \quad x \geq 2$

e $f(1) = 0$

$f(x) = \sqrt{x^4 - x^2} - x^2 + 1$

DOMINIO E POSITIVITÀ

Dominio:

$x^4 - x^2 \geq 0$

Risolviamo per sostituzione

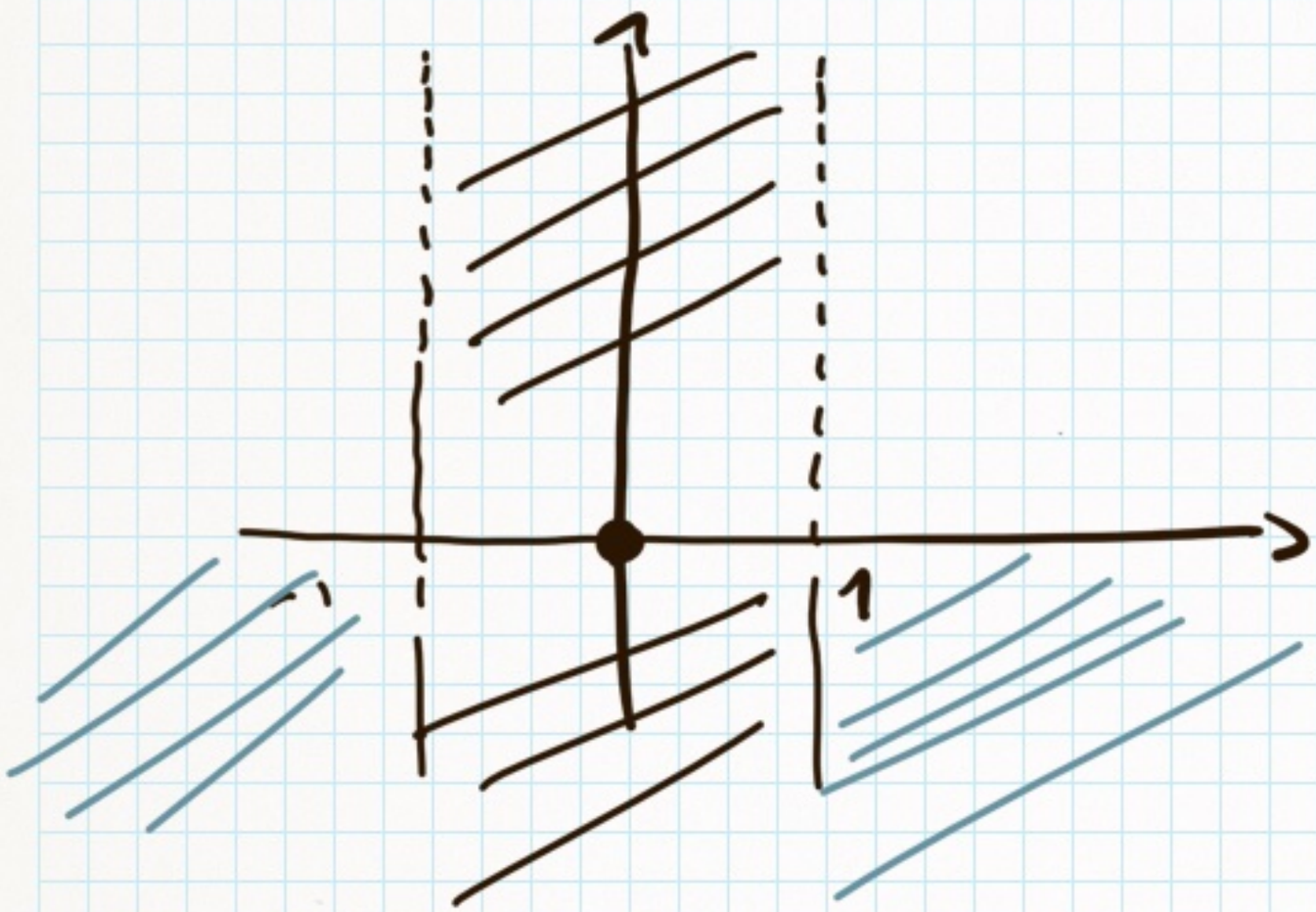
$t = x^2$

$t^2 - t \geq 0 \iff t \leq 0 \vee t \geq 1$

Torniamo indietro, $t = x^2$

$$x^2 \leq 0 \quad \vee \quad x^2 \geq 1$$

$$\boxed{x = 0} \quad \vee \quad x \leq -1 \vee x \geq 1$$



Passiamo a studiare la positività

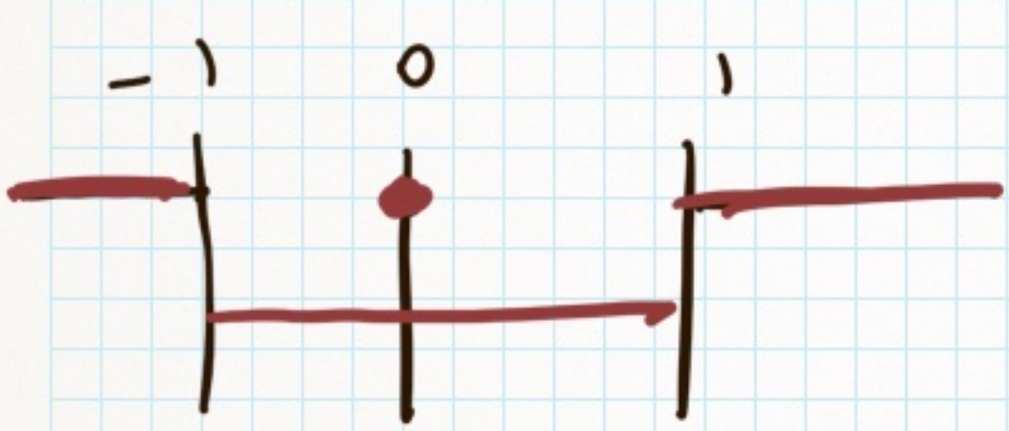
$$\sqrt{x^4 - x^2} \geq x^2 - 1$$

$$\sqrt{x^4 - x^2} \geq x^2 - 1$$

$$\begin{cases} x^4 - x^2 \geq 0 \\ x^2 - 1 < 0 \end{cases}$$

① $x \leq -1 \vee x = 0 \vee x \geq 1$ (giati fainetai)

$-1 < x < 1$



$$\vee \begin{cases} x^2 - 1 \geq 0 \\ x^4 - x^2 \geq (x^2 - 1)^2 \end{cases}$$

②

$$\boxed{x = 0}$$

② $\begin{cases} x \leq -1 \vee x \geq 1 \\ x^2 - x \geq x^2 - 2x + 1 \end{cases}$

$\begin{cases} x \leq -1 \vee x \geq 1 \\ x^2 - 1 \geq 0 \end{cases}$ $\begin{cases} x \leq -1 \vee x \geq 1 \\ x \leq -1 \vee x \geq 1 \end{cases}$

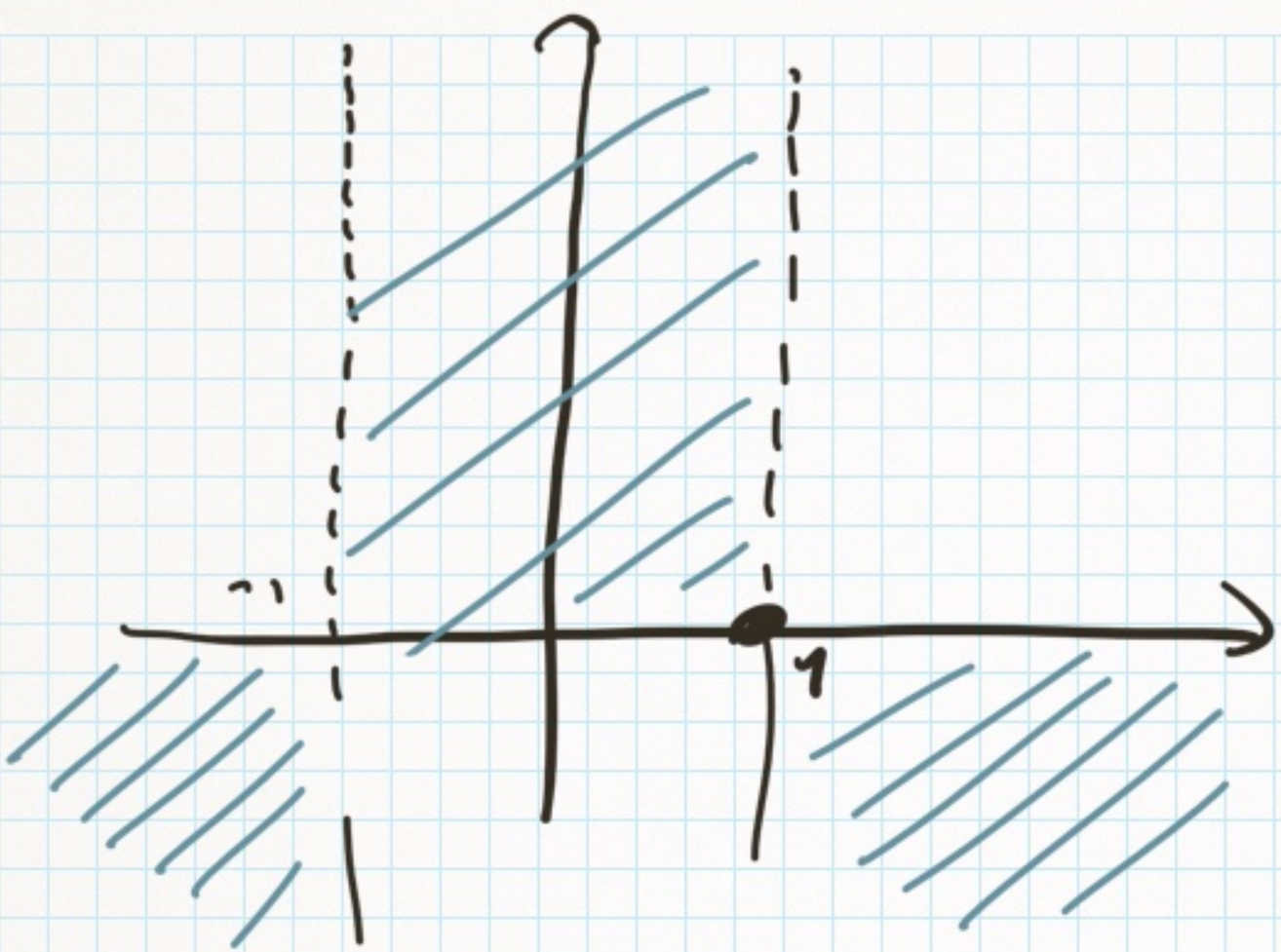
$x \leq -1 \vee x \geq 1$

$x = 0$

$f(x) = 2^{\frac{x-1}{x+1}} - 1$

Studiare dominio e positività.

$x \neq -1$



$f(x) \geq 0$

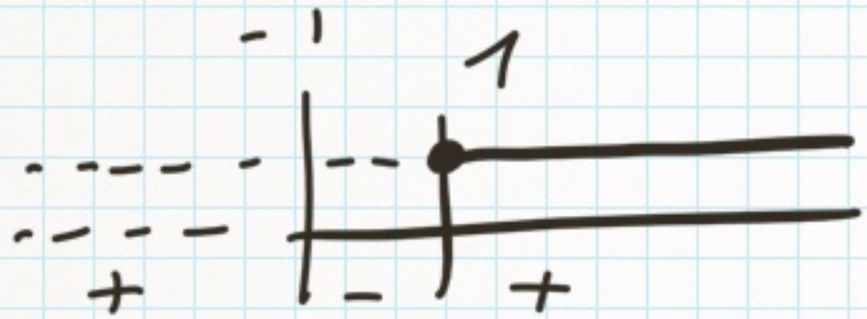
$$2^{\frac{x-1}{x+1}} \geq 1 = 2^{0}$$

$$\log_2 2^{\frac{x-1}{x+1}} \geq \log_2 1 = 0$$

$$\frac{x-1}{x+1} \cdot \underbrace{\log_2 2}_{\| \; 1} \geq 0 \Rightarrow \frac{x-1}{x+1} \geq 0$$

$x-1 \geq 0 \qquad x \geq 1$

$x+1 > 0 \qquad x > -1$



$x < -1 \quad \vee \quad x \geq 1$ e $f(x) = 0$

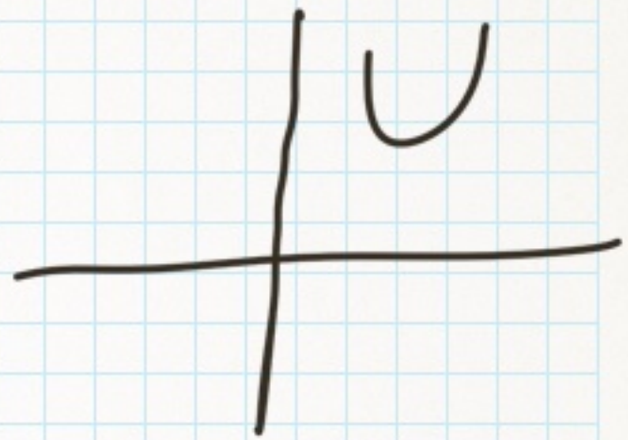
Determinare dominio e positività di:

$$f(x) = \arcsin(\underbrace{x^2 - x})$$

Dominio $-1 \leq x^2 - x \leq 1$

$$\begin{cases} x^2 - x \geq -1 \\ x^2 - x \leq 1 \end{cases}$$

$$\begin{cases} x^2 - x + 1 \geq 0 & (1) \\ x^2 - x - 1 \leq 0 & (2) \end{cases}$$



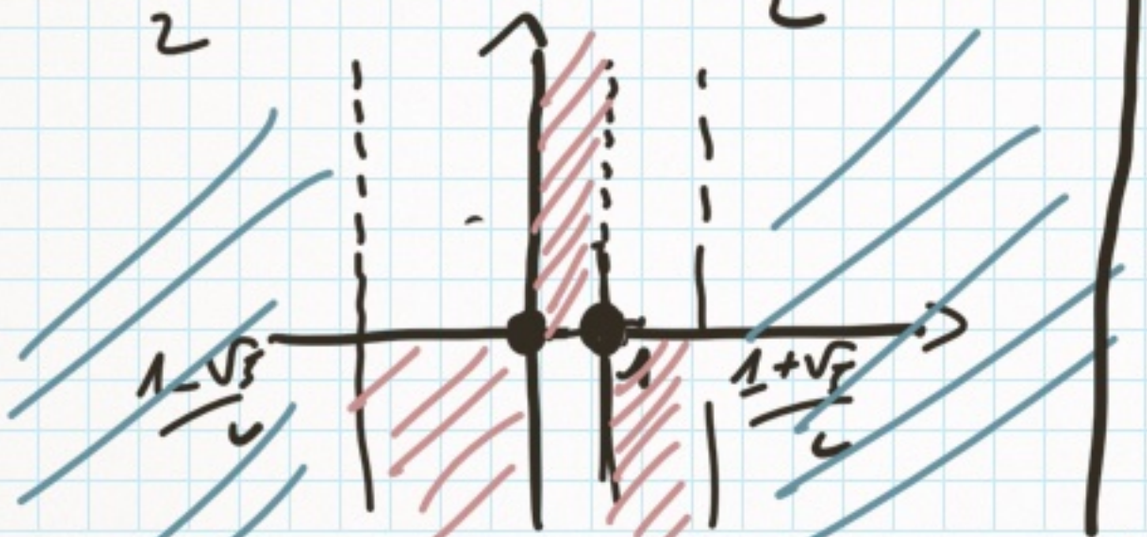
(1) $x^2 - x + 1 \geq 0 \qquad \forall x \in \mathbb{R}$

$\Delta = 1 - 4 = -3 < 0$

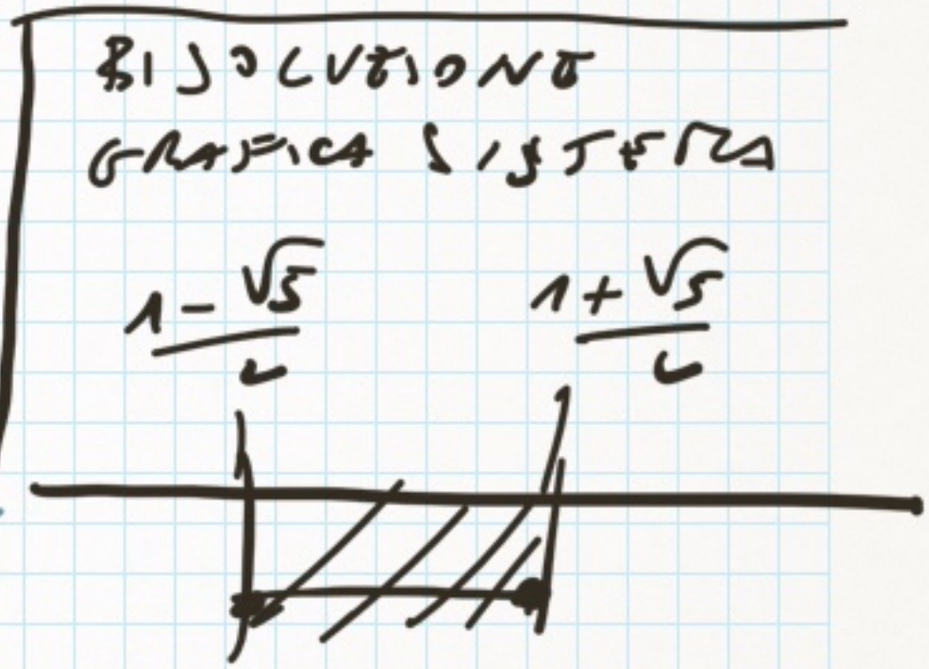
(2) $x^2 - x - 1 \leq 0 \iff \frac{1-\sqrt{5}}{2} \leq x \leq \frac{1+\sqrt{5}}{2}$

$x^2 - x - 1 = 0$

$\frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$



SOLUZIONE GRAFICA SISTEMA



Positività

$f(x) \geq 0$

$x^2 - x \geq 0$ $\quad x \leq 0 \ \vee \ x \geq 1$ (che sono nel dominio)

$x^2 - x = 0 \iff x = 1 \ \vee \ x = 0$

DEVO RICORDARE DI INTERSECARE (ANCHE SOLO GRAFICAMENTE CON IL DOMINIO)